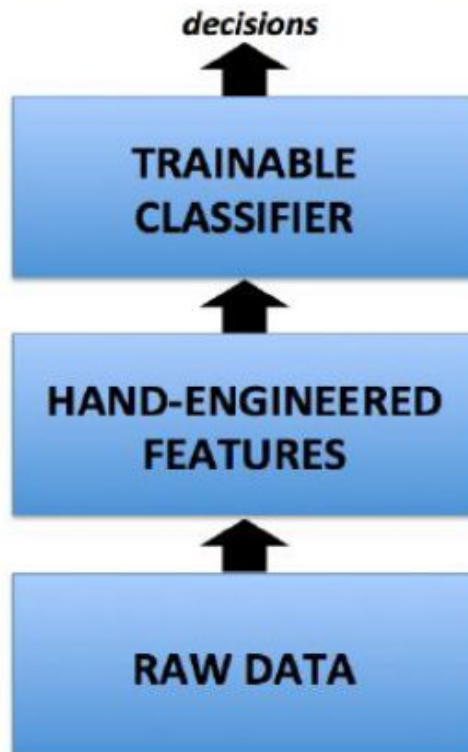


What is Deep Learning?

- A family of methods that uses deep architectures to learn high-level feature representations

STANDARD PROCESS IN MACHINE LEARNING



DEEP LEARNING



Vanishing Gradient

- Back propagation algorithm works by going from the output layer to the input layer, propagating the error gradient on the way.
- Gradients often get smaller and smaller as the algorithm progresses down to the lower layers.
- As a result the gradient descent updates in the lower layer weights virtually unchanged.
- This makes training never converges to a good solution.
- This is called vanishing gradient.

Exploding Gradients

- Some times the gradients can grow bigger and bigger, so many layers gets large weight updates and the algorithm diverges.
- This is called exploding Gradients, which is popularly seen in recurrent neural networks.

Gradient clipping

- A popular technique to lessen the exploding gradients problem is to simply clip the gradients during backpropagation so that they never exceed some threshold (this is mostly useful for recurrent neural networks)